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**Autonomous Visual Docking of a BlueROV2:
Simulation Terminal Guidance and Empirical
Limits of ArUco Perception Underwater**

Thesis submitted as a requirement for the degree of Bachelor of
Engineering

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ORIGINALITY STATEMENT

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Abstract

This thesis develops the optical terminal stage of a vision-based underwater docking system for a low-cost ROV. A perception–guidance–control pipeline using position-based visual servoing is shown in simulation to dock a BlueROV2 autonomously onto a swaying dock through coarse-approach and fine-alignment phases. In parallel, the operating envelope of passive ArUco perception is characterised from real underwater imagery: detection range, pose error, and degradation with distance and turbidity. Together these bound the realistic operating range of marker-based terminal docking and identify where a complementary far-field cue becomes necessary.

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1 Introduction

2 Literature Review

2.1 Underwater docking: motivation and phases

2.2 Acoustic and electromagnetic guidance

2.3 Vision-based docking and fiducial markers

2.4 Visual servoing and guidance for docking

Visual servo control divides into image-based schemes, which regulate feature error directly in the image plane, and position-based schemes, which reconstruct a metric relative pose and regulate it in Cartesian space [2]. When the perception layer already produces a full 6-DOF pose from fiducial markers of known geometry [3], position-based servoing costs nothing extra and keeps the docking geometry (standoff points, entry axes, capture tolerances) expressible in physical units, which is why it is adopted here.

A gap emerges around dock *motion*. Deployed and proposed resident docking stations are moored or suspended structures, and recent dynamic analyses show they respond measurably to wave and current forcing through their moorings [4]. Yet the docking-control literature overwhelmingly treats the dock as fixed, with environmental disturbance modelled at most as a steady current on the vehicle; the case of a dock whose pose oscillates at wave frequencies, and of estimating and tracking that motion from onboard vision alone, is comparatively unexplored. Tracking a moving target from a moving observer additionally raises a state-estimation question: a constant-velocity target model driven by white process noise [1], implemented here as an error-state Kalman filter [8], assumes measurement errors uncorrelated with the observer’s own motion, an assumption a vehicle-mounted camera can violate. Part A of this work quantifies both the envelope of docking against a moving dock and the consequence of that violated assumption.

2.5 Simulation of underwater vision and docking

2.6 Research gap

3 Methodology

3.1 System architecture

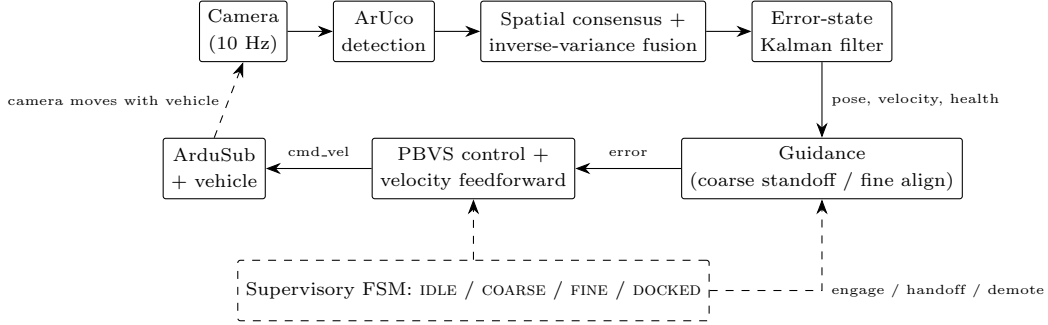


Figure 1: System architecture. Perception runs entirely in the loop; the supervisory state machine sequences the mission and gates the controllers on the filter’s health signal. The feedback path from vehicle to camera is the ego-motion coupling analysed in Section 4.1.

The system is a perception, guidance and control pipeline for the terminal stage of docking. Perception estimates the dock pose from passive ArUco markers and publishes a filtered pose, a dock velocity estimate, and a semantic health signal. Guidance converts the dock pose and the vehicle pose into a body-frame error to a target point. Control regulates that error with position-based visual servoing (PBVS) and writes a body velocity command. A supervisory state machine sequences the mission through IDLE, COARSE (drive to a standoff point on the dock entry axis), FINE (align and advance into the entry) and DOCKED, with demotion paths on loss of the perception signal.

PBVS was selected over image-based servoing because the marker pipeline already produces a full metric 6-DOF pose: fusing multiple markers of known geometry yields a calibrated dock pose, so servoing in pose space costs nothing extra and keeps the guidance geometry (standoff points, entry axis, capture tolerances) expressible in physical units [2].

The scope boundary is the optical terminal stage: trials begin with the dock inside the camera field of view at several metres range. Long-range acquisition (acoustic homing or an active light cue) is assumed to have delivered the vehicle to this point and is out of scope.

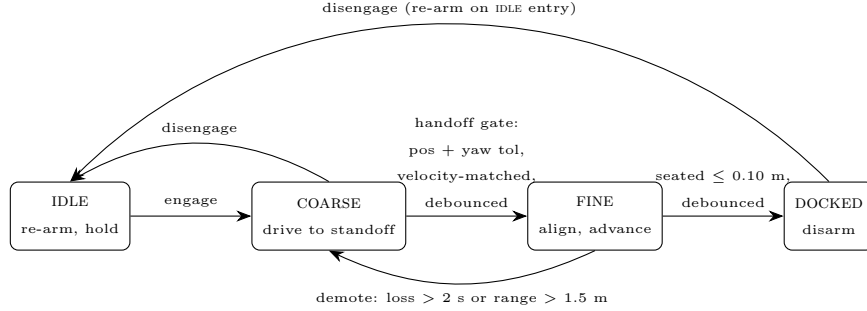


Figure 2: Docking state machine with the transition gates annotated. Demotion paths make perception loss recoverable rather than terminal.

3.2 Part A: Docking simulation

3.2.1 Simulation environment

The simulation runs ROS 2 Jazzy with Gazebo (gz-sim 8, DART physics) and an ArduSub software-in-the-loop autopilot coupled through the ArduPilot Gazebo plugin, using the BlueROV2 Heavy model with hydrodynamic and buoyancy plugins from the blue-sim stack. The vehicle carries a forward camera simulated at 10 Hz. The world places a docking station 5 m ahead of the vehicle spawn point at 1.5 m depth, carrying a printed board of nine ArUco markers (47.5 mm to 200 mm) whose layout matches the physical board of Part B.

Dock motion is generated by a custom Gazebo plugin that drives the dock through a sinusoidal sway and heave of amplitude 0.1 m with a 90-degree relative phase, producing an orbital motion representative of wave-driven excitation of a moored structure. The period is configurable per launch (18 s nominal; 6 s to 20 s in the envelope sweep) and the plugin publishes the true dock pose as a ground-truth channel used only for scoring, never by the vehicle. Mechanical capture and contact are not modelled: success is defined at the capture envelope (Section 3.2.4) and would-be contacts are scored post hoc from ground truth.

The swept period band and amplitude are anchored to real ocean forcing. Surface gravity waves occupy roughly 5 to 20 s: locally generated wind sea at 5 to 8 s and swell from 8 s up to about 20 s [5], and the long-term mean off the New South Wales coast, the operating context of this work, is a significant wave height of 1.6 m at a mean period of 8 s [7]. A submerged dock, moored or suspended [4], is forced by the wave orbital motion, whose amplitude decays with depth as e^{-kz} with $k = 2\pi/L$ and deep-water wavelength $L = gT^2/2\pi \approx 1.56 T^2$ [5], so long-period swell penetrates far deeper than wind sea. Taking the water-particle orbital excursion as a proxy for the excursion of a compliantly moored dock, the simulated 0.1 m amplitude corresponds, in the mean NSW sea state, to a dock at approximately 19 m depth for the 6 s period, 33 m at 8 s, 52 m at 10 s and beyond 200 m at 20 s; shallower docks see larger amplitudes, deeper docks only the long-period components. Sweeping 6 to 20 s at a fixed 0.1 m amplitude therefore isolates the period dependence while keeping every cell identifiable with a physically realistic sea state and operating depth (the deep-water decay is conservative for the longest periods at continental-shelf depths, where finite-depth theory retains still

more motion).

3.2.2 Perception (simulated ArUco pipeline)

The full perception pipeline runs in the loop; the controller never sees ground truth. Markers are detected on the simulated camera stream and each detection is converted to an implied dock-origin pose using the known marker layout. A spatial-consensus test rejects outlier detections, and the surviving markers are fused by inverse-variance weighting with a range-scaled noise model, $\sigma = \alpha r^2/s$ for a marker of size s at range r , so distant small markers contribute less than near large ones.

The fused measurement feeds a nine-state error-state Kalman filter over dock position, dock velocity and a small-angle rotation error carried about a reference quaternion [8]. Two process-noise regimes are defined. The *static* regime pins the velocity states (zero initial velocity uncertainty and zero velocity process noise), which is algebraically equivalent to a constant-position filter and serves as the baseline. The *sway* regime models constant-velocity motion with a white-noise-acceleration process of density σ_a (0.16 m/s² nominal, exposed as a parameter), enabling dock velocity estimation and the feedforward of Section 3.2.3. Updates are gated by a Mahalanobis test, the velocity state is bounded by a physical dock-speed limit (0.2 m/s), and the filter publishes its velocity state as a twist alongside the pose.

A health classifier maps filter state to WARMING, HEALTHY, DEGRADED or STALE from the age of the last accepted measurement and the position uncertainty: healthy requires a recent update and position standard deviation under 0.04 m, and the estimate is declared stale when either the measurement age exceeds 3 s or the uncertainty exceeds a 0.15 m ceiling, so controllers hold rather than act on an unconstrained estimate. Downstream controllers consume only this semantic signal.

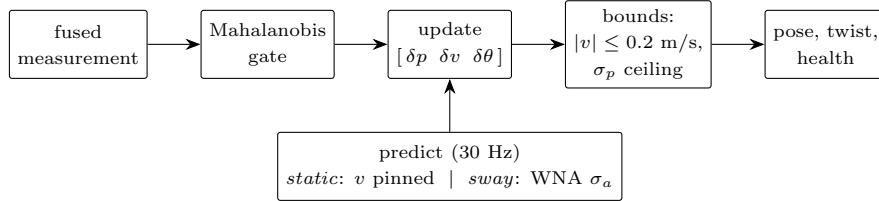


Figure 3: Filter structure: the nine-state error-state filter with the two process-noise regimes (the Part A ablation), the measurement gate, and the physically motivated state bounds.

3.2.3 Guidance and control

Coarse approach regulates a standoff point 1 m out along the dock entry axis with per-axis PD control, a distance-gated surge cap that bounds the worst-case closing speed, and the autopilot in depth-hold. Handoff to fine alignment is gated by position, boresight-offset and yaw tolerances plus a velocity-match term (an exponentially smoothed derivative of the standoff-relative position, threshold 0.15 m/s) sustained through a debounce, so the handoff cannot latch at a sway zero crossing where the position error is momentarily small while the dock moves fastest.

Fine alignment implements align-then-advance: surge toward the entry is inhibited until lateral, vertical and yaw errors are within alignment tolerances, with the autopilot in stabilise mode so the controller owns depth. The seat criterion (definition of docked) is a debounced 0.10 m range to the entry with 0.03 m lateral and vertical tolerances; the 0.10 m value is derived from the measured sensing

envelope in Section 4.1, which shows marker visibility collapsing below approximately 5 cm, so the seat is confirmed where vision is still reliable and the final centimetres are delegated to the capture mechanism assumed by the envelope definition.

Both controllers accept a dock-velocity feedforward built from the filter velocity state rotated into the body frame. The feedforward is clamped dock-relatively (the feedback term retains its full authority budget and the feedforward is limited separately at 0.25 m/s), scaled by a distance-gated authority ramp (zero beyond 1.5 m from the standoff, full inside 0.3 m), and converted from physical velocity to command units by a gain of 0.6 identified from the plant response: the command channel is effort-like, with a measured steady gain of 1.3 to 1.6 and approximately 240 ms of lag between command and vehicle velocity. In the static filter regime the velocity estimate is identically zero, so the feedforward vanishes and the system reduces exactly to the baseline controller; this is the ablation used throughout Part A.

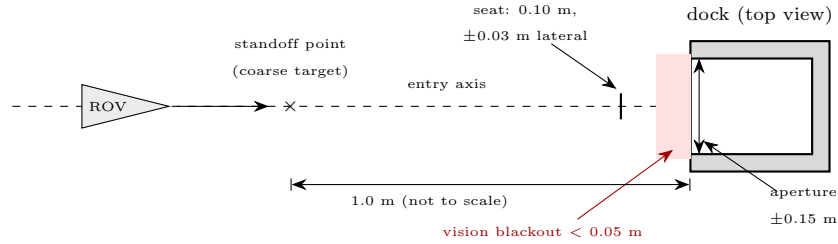


Figure 4: Terminal docking geometry (top view, not to scale): the coarse standoff point, the fine-alignment entry axis, the 0.15 m scoring aperture, the 0.10 m seat criterion, and the close-range vision blackout that motivates seating before the blind zone.

3.2.4 Experimental method

Trials are executed by an unattended harness that launches the full stack, waits for the filter to initialise and the autopilot mode to settle, engages the state machine, records a curated set of topics, polls for the docked state with a timeout, and tears the stack down between trials. Recording runs in a low-impact configuration after measurements showed recorder load itself degrading the perception cadence it was capturing.

The envelope experiment sweeps filter regime (*static* baseline versus *sway* method) by dock sway period {20, 16, 12, 10, 8, 6} s with three sway start phases per cell ($n=3$) plus a static-dock sanity cell per regime, 38 trials per sweep. Scoring uses the ground-truth channel only. Each trial receives one of three outcomes: CLEAN DOCK (docked with no pre-latch off-aperture crossing), CONTACT (the vehicle volume crossed the dock structure envelope more than 0.15 m from the calibrated entry axis before the seat latched; the trial is censored at that instant because subsequent motion traverses space a physical dock would deny), or NO DOCK. Crossing the entry plane inside the aperture is normal insertion and is not a contact. The aperture radius is a stated assumption standing in for a capture funnel; the two observed populations (centred insertions at 0.003 to 0.05 m offset versus misses at 0.17 m and beyond) are far from the boundary, so conclusions are insensitive to its exact value. The full sweep was run twice, on two hosts (real-time factors 0.6 and 1.0) and two filter revisions, to check that conclusions do not depend on load, hardware or incidental tuning.

3.3 Part B: Real-CV validation

3.3.1 Marker target and its as-printed geometry

The target uses the original ArUco dictionary (5×5 payload) with the dock’s marker set: two 155.64 mm markers (IDs 201, 202), one 77.82 mm (301), four 46.69 mm (302 to 305) and two 36.96 mm (401, 402), printed across three A4 sheets taped to a rigid panel. The as-printed geometry was estimated from the data rather than assumed: a size-weighted least-squares fit over all nine markers (residuals under 0.4 mm) shows the sheets were printed at 95.9% of nominal scale, so the analysis uses the measured black-square sides (149.4, 74.7, 44.4 and 35.6 mm) throughout. The fit is corroborated independently by the 201-to-202 baseline, which matches two butted A4 pages plus millimetres of tape gap only under the corrected scale; using nominal sizes would silently inflate every derived range by 4.1%. The as-measured layout is shown in Figure 5.

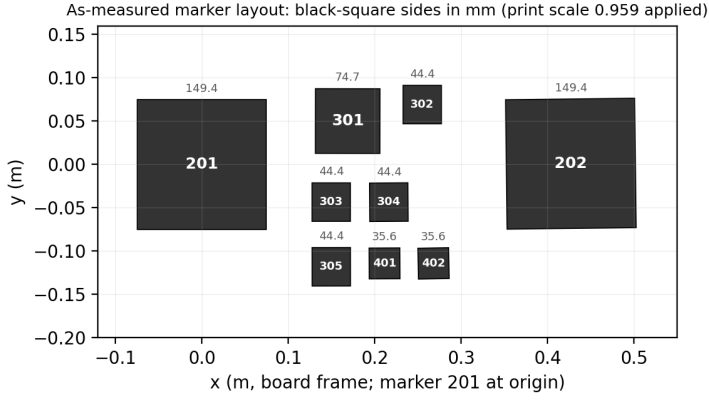


Figure 5: As-measured marker layout in the board frame, positions and black-square sides estimated from the data with the 0.959 print scale applied. The four size groups placed at a common range are the marker-size lever of the experiment.

3.3.2 Capture setup

Data was captured in a single pool session with a ZED stereo camera, of which only the left, rectified stream was recorded (960×540 at nominal 15 Hz), giving a monocular ArUco problem; images, camera info and the ZED IMU were logged. Two runs were recorded: an on-axis walk-back (677 frames at an effective 2.5 Hz over 271 s) and an oblique walk-back from roughly 4 m of lateral offset (737 frames over 85 s, with about 42% of frames dropped in bursts by disk bandwidth; the two runs are therefore analysed separately and never pooled). The vehicle was hand-held and paced backwards from the board to beyond the detection limit, with a yaw turn at each step.

Camera intrinsics are the shipped in-air factory rectification ($f_x=f_y=797.5$, zero distortion on the rectified stream). The in-water validity of this choice was tested by a joint bundle adjustment over 1400 marker observations in 317 frames, solving for underwater focal lengths, the coplanar board layout and per-frame poses with scale pinned by the ruler-measured marker sizes: the aspect ratio solves to $f_y/f_x = 0.996$ and the isotropic optimum is only 1.9% better than the in-air value, while the flat-port refraction correction of $1.333\times$ is rejected outright (44% worse residual). The shipped intrinsics are

therefore used unchanged, and all metric ranges carry an approximately $\pm 10\%$ scale uncertainty from the focal length, which dominates the error budget. For this reason the primary independent variable throughout Part B is the calibration-free *apparent marker size in pixels*; metric range is derived and reported with its uncertainty.

3.3.3 Experiment design

The two walk-backs sweep range continuously from about 0.6 m to beyond the detection limit, and marker size is swept *by construction*: four size groups (149.4 down to 35.6 mm, a $4.2\times$ lever) are present on the board simultaneously, so every frame samples all sizes at a common range and water condition. Viewing angle was not controlled: the on-axis run has a median marker incidence of 13° and the oblique run 58° , two disconnected regimes confounded with range and marker size, so no detection-versus-angle curve is reported.

No external ground truth exists (no tape or survey range, and vehicle odometry was not recorded); scale is anchored solely by the ruler-measured marker sizes and attitude solely by an IMU gravity check. Detection is scored per marker-frame trial against apparent-size bins with a minimum of ten trials per bin and Wilson confidence intervals (2551 trials on-axis, 950 oblique). Maximum detection range is established uncensored: the on-axis walk continued for a further 79 s (about 3 m) beyond the last detection with zero detections, and two independent end-of-run estimates (walk-back rate extrapolation and board apparent width in the final frame) agree at 7.8 to 8.1 m, so range limits below are detection failures, not end-of-data artefacts.

3.3.4 Turbidity measurement and synthetic degradation

Water attenuation was measured photometrically from the target itself, without dosing or a turbidity meter. A black patch (marker border) and a white patch (surrounding sheet) at the same range share an identical veiling-light term in the image formation model $I = J e^{-\beta d} + B(1 - e^{-\beta d})$, so their difference is free of B : $\text{contrast}(d) = (J_w - J_b) e^{-\beta d}$, and β is a straight-line fit in log space. Two corrections proved load-bearing: an instrument-response correction $k(\text{apparent px})$, measured at fixed range so it is independent of β , without which small (hence distant-looking) markers read up to 18% low and the bias imitates attenuation; and per-marker intercepts, since the three sheets sit under different local lighting. The underlying image formation model is the standard attenuation-plus-veiling form [6].

Higher turbidity than the pool's was synthesised by re-attenuating each real frame in place: $I' = B + (I - B) e^{-\Delta\beta d(x)}$ per channel and per pixel, with the per-pixel range $d(x)$ taken from the board-plane pose and $\Delta\beta = m\beta$ for multipliers $m \in \{0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8\}$, where $m=0$ reproduces the real frame. Results are reported against total optical depth $\tau = (1+m)\beta d$ rather than named water types: range enters the model only as the product βd , so any uniform scale error in d is absorbed into β , making the τ axis immune to the $\pm 10\%$ range uncertainty and portable across waters and cameras. The synthesis adds no scattering blur, so any benefit a contrast-enhancing preprocessor shows on synthesised frames is an upper bound. The full chain is summarised in Figure 6.

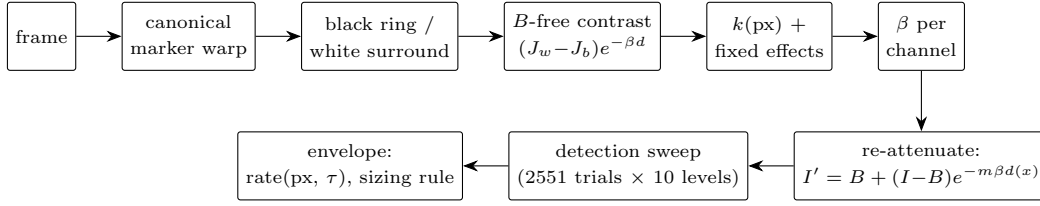


Figure 6: Turbidity processing: the top row measures the water’s attenuation from the target itself via the veiling-free contrast; the bottom row stresses the same real frames to higher optical depth and maps the detection envelope.

3.3.5 Detection pipeline and metrics

Detection uses OpenCV 4.10 with the current `ArucoDetector` API, subpixel corner refinement, and `adaptiveThreshConstant` lowered from the default 7 to 3, the best of a grid over $\{3, 5, 7\}$ scored at the highest synthesised turbidity (selected without a holdout, so its reported rate is an optimistic bound). Single-marker pose, the quantity under test, uses planar PnP (IPPE square); the reference for self-consistency is a multi-marker board PnP computed leave-one-out, with the marker under test excluded from its own reference. Off-board IDs are deliberately not filtered so the misidentification rate remains measurable. Reported metrics are detection rate per apparent-size bin and per derived-range bin, uncensored maximum detection range per marker, single-marker translation and rotation deviation from the board reference (self-consistency, not accuracy, since no ground truth exists), misidentification rate, and an IMU gravity check as the sole camera-independent attitude validation.

4 Results and discussion

4.1 Part A: Simulation results

4.1.1 Perception envelope

The filter’s accuracy is governed by range through the $\alpha r^2/s$ measurement noise model. With the vehicle stationary 5 m from the dock the fused noise floor is approximately 0.11 m and the filter reports DEGRADED throughout; at 2.2 m the floor drops to approximately 0.02 m, the filter is HEALTHY for 92% of the run (median position standard deviation 0.018 m), and the velocity state tracks the true dock velocity with amplitude ratio 0.85 and correlation 0.89 (Figure 7). At the opposite extreme, marker visibility collapses at contact range: binning fine-alignment health by ground-truth range shows 84% to 100% healthy beyond 5 cm of the entry but only 21% inside it, a blackout produced by the large markers overflowing the camera field of view. This measured envelope, reliable from roughly 2.5 m down to 0.05 m, motivates both the 0.10 m seat criterion and the delegation of the final centimetres to mechanical capture.

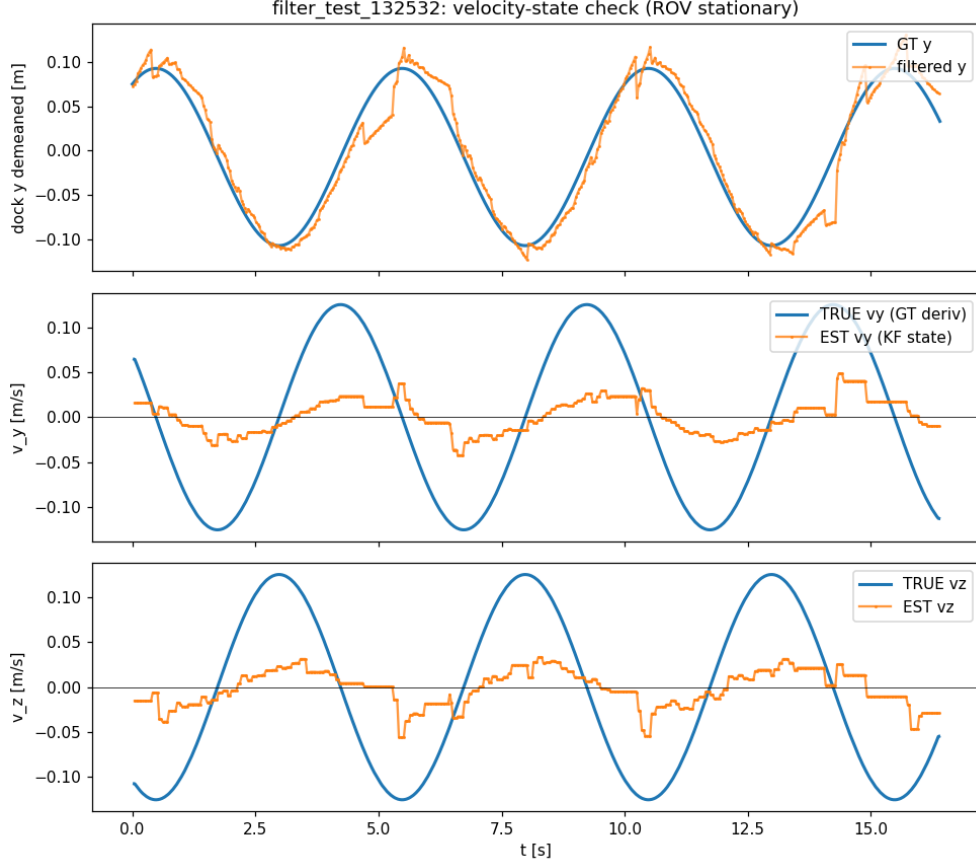


Figure 7: Dock velocity estimation with the vehicle stationary at 2.3 m (18 s sway). The velocity state (orange) tracks the ground-truth dock velocity (blue) at 0.85 amplitude with correlation 0.89.

4.1.2 Docking demonstration

Against the nominal 18 s, 0.1 m orbital sway, the system docks autonomously end to end: engagement to coarse at $t+3$ s, handoff to fine alignment at $t+14$ s, seated and docked at $t+25$ s with no demotions, latching 7 to 8 cm from the entry and coming to rest 4 cm in front of the dock plane. This trial, and every sweep trial below, ran the full perception pipeline in the loop with no ground-truth input to the vehicle.

4.1.3 Docking envelope versus sway period

Table 1 and Figure 8 give the headline result of the sweep ($n=3$ per cell, contact-censored scoring). The baseline (static filter, no feedforward) docks in every trial at every period in the raw scoring, and cleanly down to 8 s; at 6 s, two of three trials brush the aperture boundary at low speed (0.03 to 0.06 m/s) before latching. The method (constant-velocity filter with feedforward) matches the baseline down to 12 s, degrades at 10 s (one pre-latch contact), and fails at 8 s and 6 s; its single raw success at 8 s was contact-assisted.

Table 1: $P(\text{dock})$ by sway period and scoring, $n=3$ per cell. Clean excludes trials with a pre-latch off-aperture crossing.

Period [s]	Baseline (ff off)		Method (CV + ff)	
	raw	clean	raw	clean
20	1.00	1.00	1.00	1.00
16	1.00	1.00	1.00	1.00
12	1.00	1.00	1.00	1.00
10	1.00	1.00	1.00	0.67
8	1.00	1.00	0.33	0.00
6	1.00	0.33	0.00	0.00

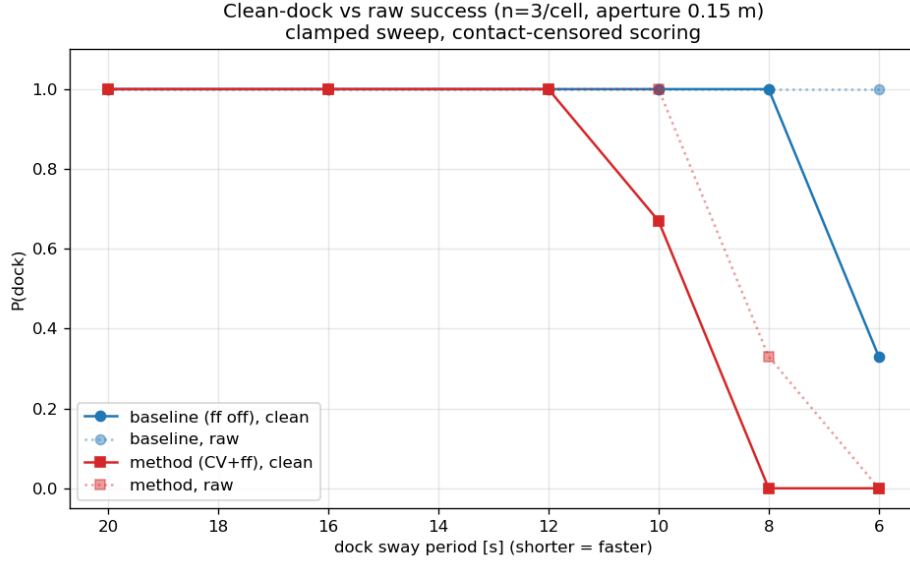


Figure 8: Docking success versus sway period under raw (dotted) and contact-censored clean (solid) scoring.

In the operational terms of Section 3.2.1, each period cell maps to a sea state and dock depth. The mean NSW sea (8 s period, 1.6 m significant height) forces a 0.1 m sway on a dock near 33 m depth, which the baseline handles cleanly; the 6 s wind-sea cell corresponds to a shallow dock near 19 m in the same sea, where the baseline still docks in every trial under raw scoring; and the longer-period cells correspond to deeper moorings reached only by swell, where both configurations succeed. The measured envelope therefore covers moored docks across typical continental-shelf operating depths, with residual risk concentrated in shallow, short-period deployments, and the method’s cliff at 8 to 10 s sits at the region’s mean wave period, which is precisely why the baseline’s structural immunity matters operationally.

Clean insertions cross the entry plane 0.003 to 0.05 m off axis at 0.06 to 0.11 m/s closing speed, and all observed contacts occur below 0.06 m/s. These figures form the interface requirement for a capture mechanism: a funnel with a capture radius of at least 0.15 m and tolerance for approximately

0.1 m/s closing speed captures everything the baseline delivers down to 8 s periods.

4.1.4 Failure analysis: ego-motion instability of the velocity estimator

The method’s fast-period failures trace to a single mechanism. The dock measurement is reconstructed in the world frame through the transform of a moving camera, so a fraction of the vehicle’s own motion leaks into the measured dock position. This leakage is *correlated with the vehicle state*, violating the filter’s white-noise assumption. The static filter, with no velocity state, absorbs it as position noise and remains faithful (estimated sway amplitude 0.7 to 0.9 times truth in all baseline runs). The constant-velocity filter integrates it into a phantom dock velocity and extrapolates: estimated amplitude reaches 2.4 to 5 times truth in successful method runs and 18 to 52 times truth in the failures, where the estimate has decoupled from the real dock (correlation 0.00 with the true dock while the estimate’s excess motion correlates at 0.88 with the vehicle’s own position). The feedforward closes the loop by injecting the phantom velocity directly into the command; with the identified plant lag of approximately 240 ms, the loop gain crosses unity between the 10 s and 8 s periods, which is where the measured cliff sits.

The failure is self-concealing: every alignment signal the system exposes is computed against its own estimate, and in the failed trials those signals read seat-tolerance alignment (lateral errors of 0.000 to 0.017 m) while ground truth shows the vehicle 0.17 to 0.48 m off axis at the dock plane. Only the ground-truth channel exposes the divergence, an argument for independent verification in any real deployment.

Three progressively stronger mitigations were evaluated. Bounding the velocity state at the physical dock speed and adding the covariance ceiling eliminated the metre-scale runaway and the resulting collisions, but did not restore fast-period docking. Sweeping the process noise density σ_a over 0.02 to 0.16 m/s² found no stable setting: estimated amplitude remained 5 to 37 times truth at every value with no monotonic trend, because a smaller gain also makes the filter slower to shed an acquired phantom velocity. The baseline’s immunity is structural rather than a tuning point: its velocity gain is exactly zero, and no finite gain interpolates to that property while the measurement error remains correlated with vehicle motion. The conclusion is that dock velocity feedforward from a world-frame constant-velocity estimator requires explicit ego-motion compensation, by fusing vehicle odometry into the filter or estimating in a vehicle-relative frame, identified as future work.

4.1.5 Replication

The sweep was run in full on two hosts (real-time factors 0.6 and 1.0) and two filter revisions (before and after the velocity bounds). The baseline’s uniform success and the method’s cliff between 10 s and 8 s reproduced in both, indicating the result reflects the estimator design rather than compute load, hardware or incidental tuning.

4.2 Part B: Real-CV results

4.2.1 Detection rate versus marker size and range

Detection is governed by apparent size in pixels with a sharp floor. Across all four size groups the rate is zero below 21 px, transitional in the 21 to 28 px bin (0.21 to 0.59 depending on size), and above 0.85 from 38 px upward (Figure 9). On the derived metric axis this maps to overall on-axis rates of 0.93 for the 149.4 mm markers, 0.61 at 74.7 mm, 0.43 at 44.4 mm and 0.29 at 35.6 mm over the swept range. The oblique run (median incidence 58°) detected only the 149.4 mm markers, at 0.60 overall; the three smaller groups produced zero detections in 791 trials, indicating that oblique viewing costs roughly one size class even before the angle reaches the detector's geometric limits.

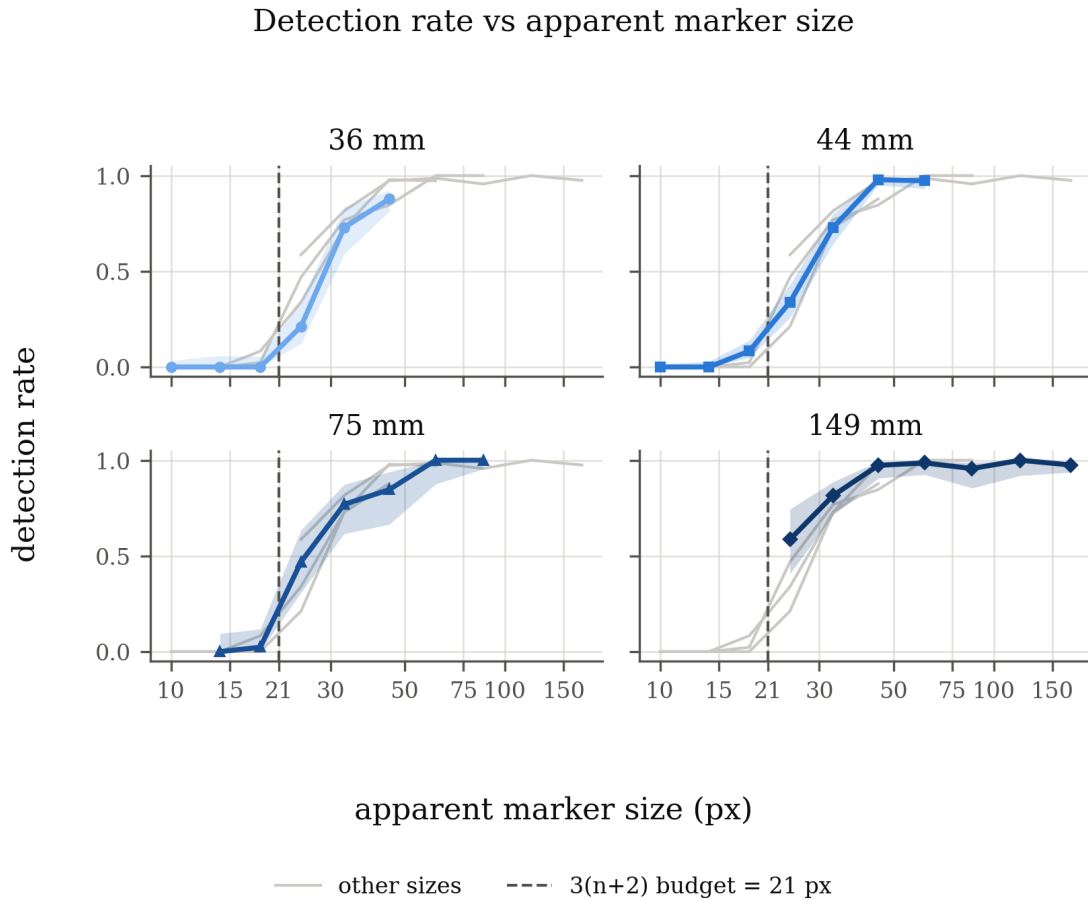


Figure 9: Detection rate versus apparent marker size (pixels), per marker size group, with Wilson 95% intervals; bins with fewer than ten trials suppressed. The axis is calibration-free.

4.2.2 Maximum detection range

Uncensored maximum detection ranges scale linearly with marker side: 5.10 and 4.87 m for the two 149.4 mm markers, 3.05 m at 74.7 mm, 1.55 to 2.17 m at 44.4 mm, and 1.10 to 1.14 m at 35.6 mm, fitted by $\text{max range} \approx 34.9 \times \text{side}$ (Figure 10). The walk-back continued approximately 3 m past

the last detection with none recorded, so these are genuine detection limits. Two distinctions matter operationally. First, only the largest markers reach the 5 m acquisition range assumed for the coarse phase, and only marginally. Second, maximum range is a single-tail-detection statistic; the *reliable* (50%) range sits lower, at 5 m a 149.4 mm marker subtends 23.8 px, below the ~ 27 px needed for 50% detection (Section 4.2.5), which is why its maximum range of 5.1 m coexists with a 50% range nearer 4.3 m. Only the second number sizes a dock.

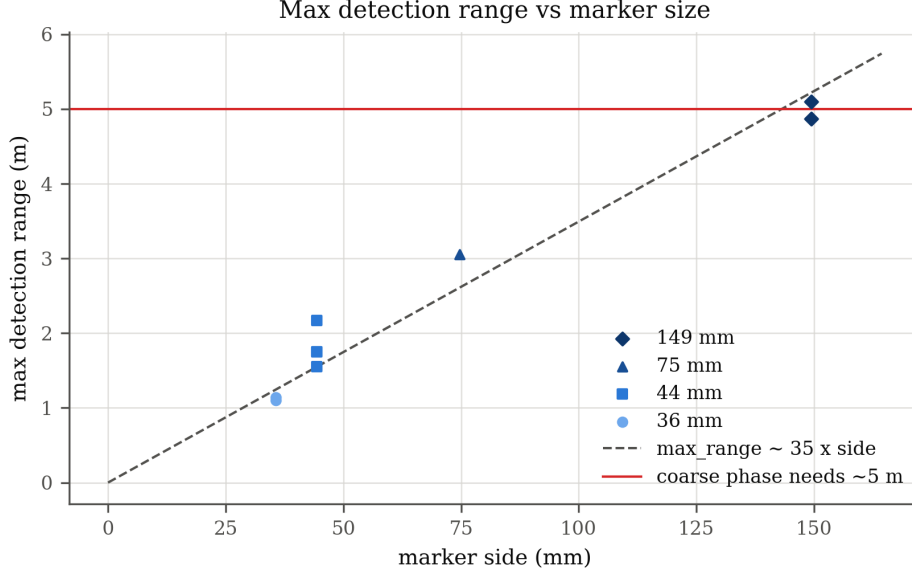


Figure 10: Maximum detection range versus marker side with the fitted linear law and the 5 m coarse-acquisition reference line.

4.2.3 Pose self-consistency and misidentification

With no external ground truth, single-marker pose is scored against a leave-one-out multi-marker board reference, a self-consistency measure, not accuracy. Translation deviation grows from 10.6 mm mean (9.4 mm jitter) in the 0.5 to 1.0 m bin to 36.1 mm mean (32.9 mm jitter) at 2.0 to 3.0 m; rotation deviation is 11 to 16° and roughly flat with range. These figures pool all marker sizes, since a per-size split would change the reference geometry with the marker under test and read backwards. The sole camera-independent validation is an IMU gravity check: over 323 frames the board’s estimated tilt from vertical has a median of 6.6°, consistent with a hand-held, approximately vertical board. A yaw-rate cross-check was not corroborative (only 7 of 25 turn segments had sufficient board coverage, and vision magnitudes did not track the gyro), and is reported as such. Misidentification was essentially absent in real water: one off-board ID in 1376 on-axis detections (rate 0.0015) and none in the oblique run.

4.2.4 Attenuation measurement

The B-free contrast fit yields, with instrument correction and per-marker intercepts over 934 samples spanning 0.62 to 3.19 m: $\beta_b = 0.279$, $\beta_g = 0.247$, $\beta_r = 0.443$, $\beta_{grey} = 0.289 \text{ m}^{-1}$ ($r^2 = 0.71, 0.67$,

0.88, 0.75; Figure 11). The strongest support for the correction chain is a quantity it never saw: the red-to-green attenuation ratio moves from 1.50 (raw) to 1.79 (corrected), against approximately 1.8 expected from water optics. A residual check over 5629 near-to-far marker pairs (not a statistical cross-validation, no holdout was used) shows the exponential form predicts real far-frame contrast to a median error of -3.1% corrected, versus $+18.5\%$ raw. At 5 m this water gives a grey optical depth $\tau = 1.44$, i.e. 23.6% of contrast surviving. The fitted veiling light B is explicitly not physically meaningful (negative r^2 ; the green estimate exceeds the white sheet’s own brightness): the method is built so that B cancels from contrast, and B enters the synthesis only as a DC term that adaptive thresholding rejects.

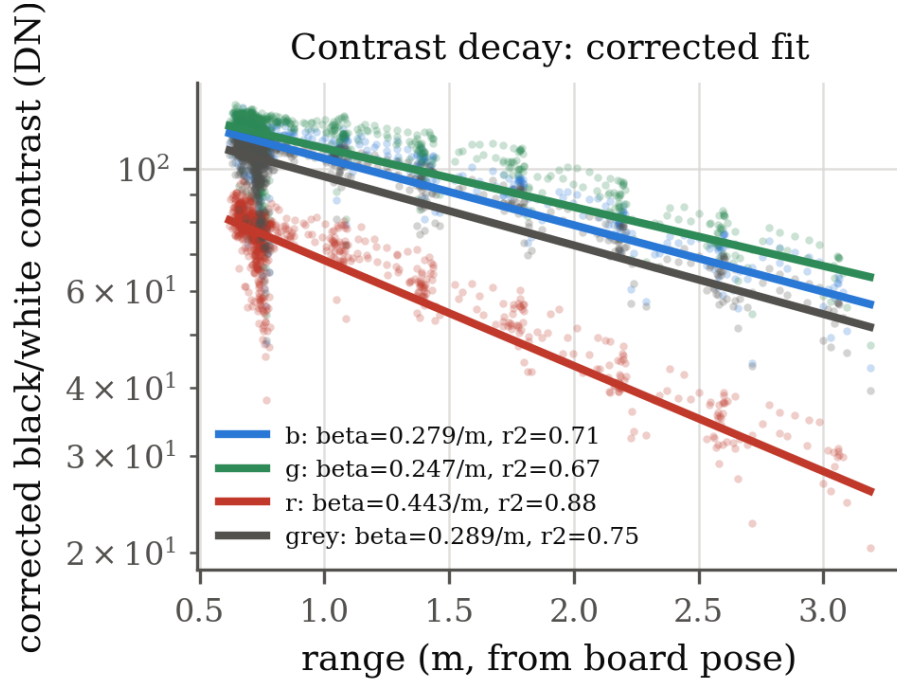


Figure 11: Instrument-corrected black-to-white contrast versus range with the fixed-effects exponential fits per channel.

4.2.5 Turbidity envelope and marker sizing

Re-attenuating all 2551 on-axis trials at ten turbidity multipliers (a fixed denominator at every level) maps the detection envelope over apparent size and total optical depth (Figure 12). The envelope has two regimes: below $\tau \approx 2$ the boundary is vertical, detection is *pixel-limited* and turbidity hardly moves it; above $\tau \approx 2.3$ the boundary is horizontal, detection collapses at every apparent size sampled, *turbidity-limited*. The pixel budget quantifies the first regime: the apparent size needed for 50% detection is 27.3 to 28.1 px, flat across a sevenfold span of optical depth (τ 0.18 to 1.30), rising only within the resolution of the binning before the collapse (Figure 13). Two conclusions follow: the textbook budget of $3(n+2) = 21$ px for a 5×5 marker underestimates the real requirement by about 30%, because it is a pure sampling argument that assumes air; and ArUco’s adaptive thresholding

makes detection remarkably insensitive to contrast loss until an abrupt threshold near $\tau \approx 2$, a cliff, not a gradient.

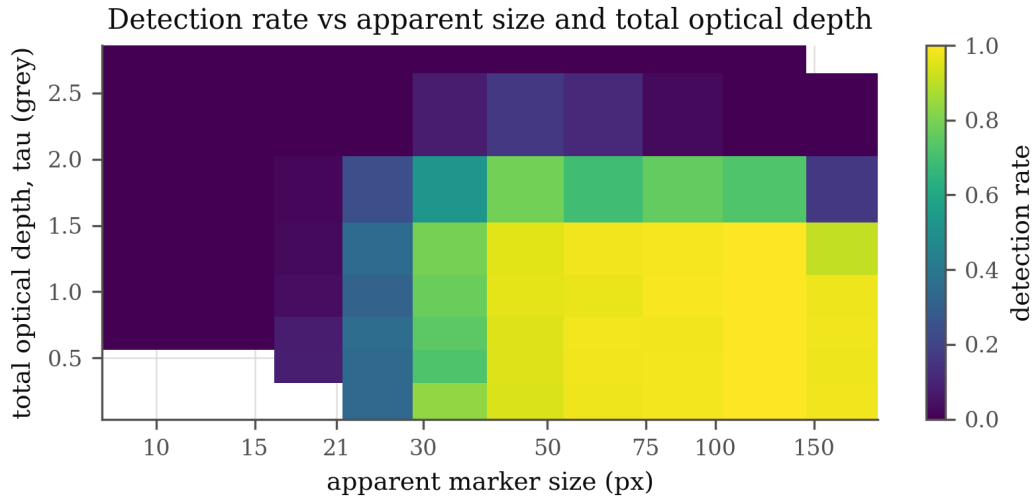


Figure 12: Detection rate over apparent size and total optical depth. The boundary is vertical (pixel-limited) below $\tau \approx 2$ and horizontal (turbidity-limited) above. White cells are physically unreachable combinations.

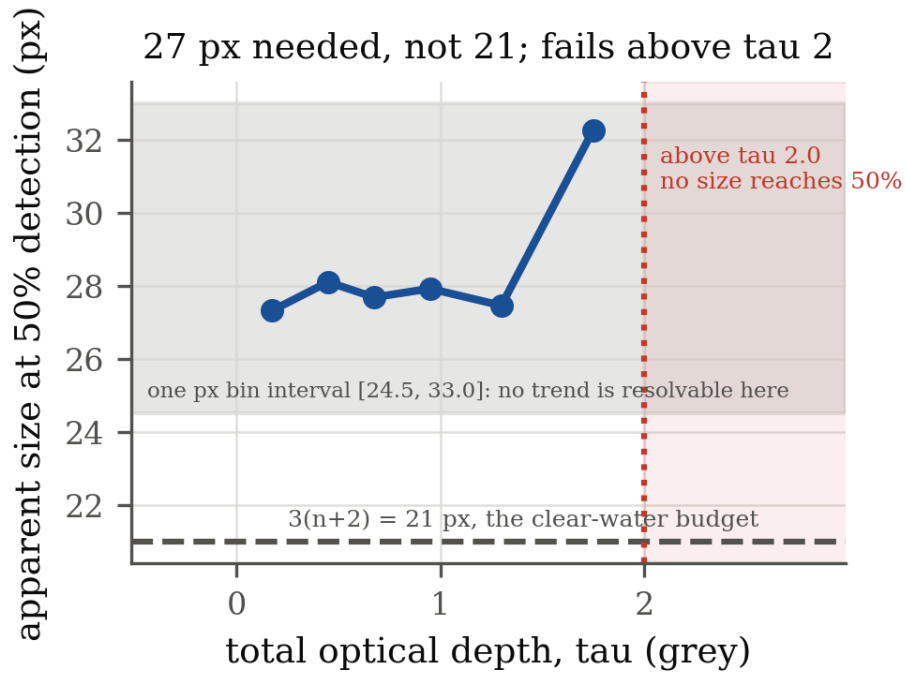


Figure 13: Apparent size required for 50% detection versus optical depth: flat at ~ 27 px across τ 0.18 to 1.30 (band shows the bin-interpolation resolution), undefined above $\tau \approx 2$ where no size sampled reaches 50%.

Inverting the pixel budget through the pinhole model sizes the dock marker: for 50% single-frame detection this pool’s water requires a 105 mm marker at 3 m and 182 mm (interval 154 to 207 mm) at 5 m. In clearer water ($\beta = 0.15 \text{ m}^{-1}$) the 5 m requirement barely drops (174 mm), confirming that below the cliff the requirement is set by pixels, not water; in murkier water ($\beta \geq 0.6 \text{ m}^{-1}$, $\tau \geq 1.8$ at 3 m) the requirement is undefined within the measured envelope, no marker size sampled achieves 50%. The textbook 21 px budget would have specified 132 mm at 5 m, 38% short in side length and roughly half the area. The synthesised degradation is illustrated in Figure 14; a forward-scatter check found no measurable edge blurring within apparent-size bins across the synthesised range (slopes scatter around zero), so at these optical depths contrast loss, not blur, is the operative mechanism.

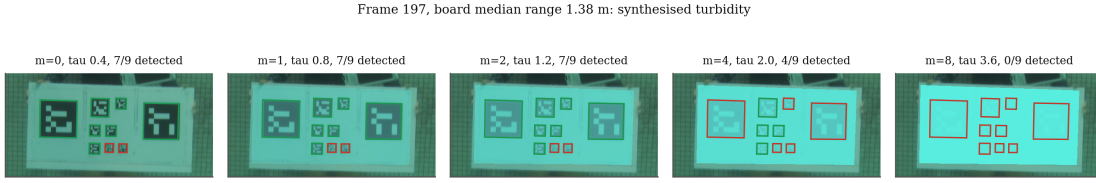


Figure 14: One frame re-attenuated at increasing turbidity multipliers with detections outlined: 7 of 9 markers at the real condition falling to 0 of 9 at the highest multiplier. The highest-multiplier panel is a diagnostic of the synthesis (its DC term is unphysical), not a depiction of real murky water.

4.2.6 Preprocessing: CLAHE evaluated and rejected

CLAHE was evaluated against the detector’s own thresholding knob over all 2551 trials at every turbidity level. In *real* water (multiplier 0) CLAHE is a net negative: detection 0.512 versus 0.519 baseline, while introducing 0.022 misidentifications per frame where the baseline has zero. Its apparent gain at high synthesised turbidity (0.337 versus 0.159 at the highest level) is measured on frames whose degradation adds no scattering noise, and is therefore an upper bound unproven for real water. Lowering `adaptiveThreshConstant` to 3 recovers about two-thirds of that gap (0.276) at a third of CLAHE’s misidentification rate (Figure 15). The engineering decision follows the real-water evidence: no CLAHE stage in the pipeline; tune the detector’s threshold constant instead.

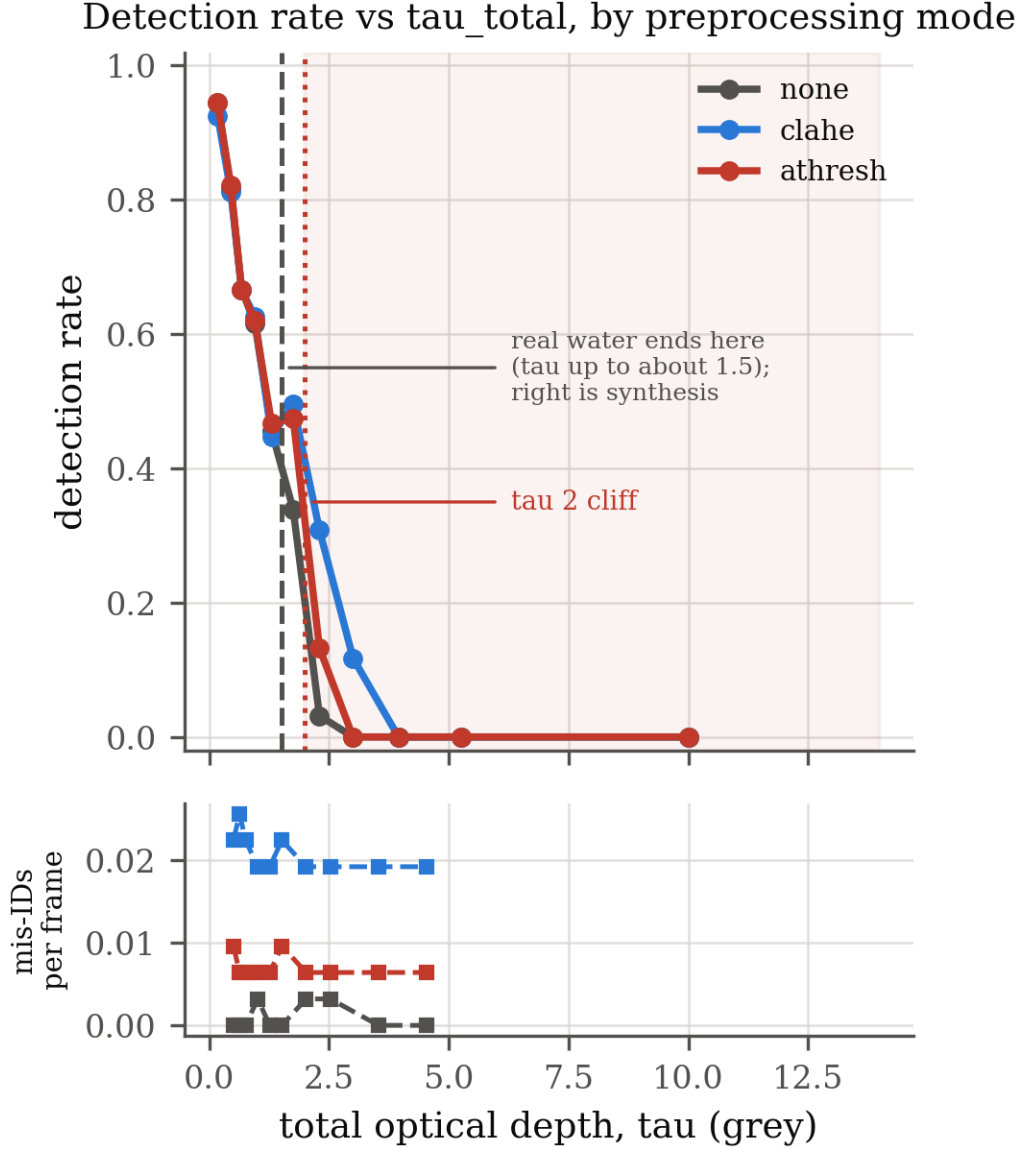


Figure 15: Detection rate (top) and misidentifications per frame (bottom) versus total optical depth for no preprocessing, CLAHE, and a lowered adaptive-threshold constant. Real water ends at the dashed line; everything beyond is synthesised.

4.2.7 Limitations

The dataset is one pool, one session: the optical-depth axis is designed to generalise (results are reported against $\tau = \beta d$), but this water is a single point and the attenuation spectrum of other waters will differ. No external ground truth exists; pose results are self-consistency only, and all metric ranges carry $\pm 10\%$ from the in-air focal length. β was fitted over 0.62 to 3.19 m, so quoting τ at 5 m assumes β constant with range. The oblique run is thin (161 detections, largest markers only) and viewing angle was never controlled. The capture was hand-held, so motion blur is present and

unmodelled, and synthesised turbidity levels derive from the same frames and are correlated samples, so the quoted confidence intervals understate uncertainty at high multipliers.

4.3 Synthesis

The two halves bound the operating envelope of passive-marker terminal docking from opposite sides. Part A bounds the *dynamics*: with the full perception pipeline in the loop, the vehicle captures a swaying dock cleanly down to 8 s periods at 0.1 m amplitude, limited by its own control bandwidth (240 ms plant lag), and delivers insertions a capture funnel of 0.15 m radius and 0.1 m/s tolerance can accept. Part B bounds the *sensing*: a marker is reliably detectable only above ~ 27 px of apparent size and below $\tau \approx 2$ of optical depth, which in the measured water prices the 5 m acquisition range at a 182 mm marker (154 to 207 mm) and rules out reliable acquisition at that range in substantially murkier water regardless of size.

The intersection is the design statement for the dock. The 200 mm-class markers assumed throughout Part A are validated by Part B as marginally sufficient for 5 m acquisition in pool-clarity water, with the per-size fusion pipeline of Part A consuming exactly the size-dependent envelope Part B measured (small markers taking over as large ones overflow the field of view at close range, whose 5 cm blackout sets the seat criterion). Beyond the measured envelope, at longer standoff or higher turbidity, the passive-marker channel provides no signal, and terminal docking must be handed a vehicle already inside the envelope by a far-field cue such as an active light beacon, which is precisely the acquisition stage this work scoped out and now quantifies the need for.

5 Conclusion

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Appendix

A. Mathematical derivations

B. Experimental run sheet

C. Additional results

D. Software and data